

CYLAB SECURITY ACADEMY

Challenge Write-Up Template

Challenge Information

Student Name	Evallo, Daniella Kim N.
Section / Class	BSIT-3G
Challenge Name	Undo
Category	General Skills
Points / Difficulty	Easy
Date Solved	08/09/2026
Flag Format Given	picoCTF{ }

1. Challenge Description

The challenge asks the player to reverse a series of Linux text transformations in order to recover the original picoCTF flag.

2. Initial Analysis / Reconnaissance

The first thing I checked was the challenge description and hint. The description says that the original flag has gone through several Linux text transformations. The hint points to the Linux `tr` command for text translation and character replacement, so I expected to reverse several simple text operations.

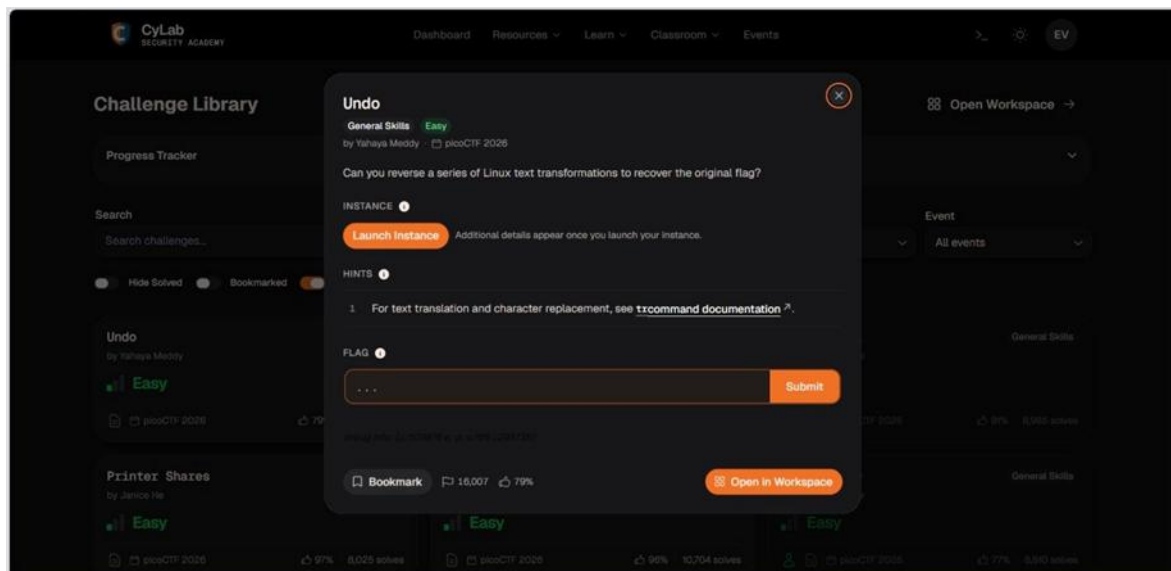
3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
<code>nc foggy-cliff.picoctf.net 65105</code>	Connect to the running challenge instance.	Opened the Text Transformations Challenge and displayed Step 1.
<code>base64 -d</code>	Decode the Base64-encoded text.	Decoded the starting string and moved to Step 2.
<code>rev</code>	Reverse the text because the hint said it was reversed.	Restored the original character order for the next step.
<code>tr '-' '_'</code>	Change dashes back into underscores.	Reversed the underscore-to-dash substitution.
<code>tr '()' '{}'</code>	Change parentheses back into curly braces.	Restored the curly braces in the flag.
<code>tr 'a-zA-Z' 'n-za-mN-ZA-M'</code>	Reverse the ROT13 transformation.	Recovered the original flag.

4. Solution Walkthrough

Step 1

I first opened the Undo challenge in the CyLab Challenge Library. I checked the challenge description, hints, and the available Launch Instance button.



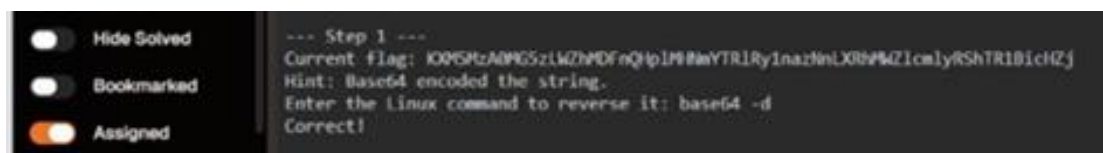
Step 2

I first opened the terminal and entered the `nc foggy-cliff.picoctf.net 65105` command provided by the challenge. After successfully connecting, the Undo challenge displayed its welcome message and instructions. It then presented Step 1, where I needed to identify and reverse the first text transformation.



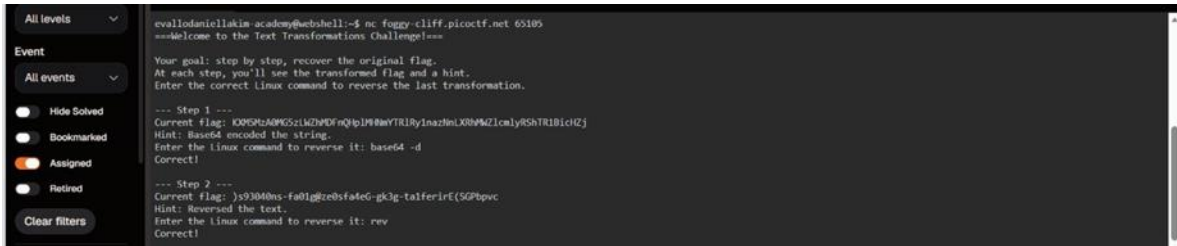
Step 3

the terminal shows that the current flag is encoded in **Base64**. The hint says that it is **Base64 encoded**, so used the ``base64 -d`` command to decode it. After run the command, the text changed and the terminal showed “Correct!”. This means It decoded it correctly and could move on to Step 2.



Step 4

The flag from the previous step was reversed. Followed the hint and used the ``rev`` command to reverse the text back to its original order. After running the command, the terminal showed the new result and **“Correct!”**. This confirmed that the Step 2 command was correct.



```
eval@kali:~/kali-academy@webshell:~$ nc foggy-cliff.picoctf.net 65105
==Welcome to the Text Transformations Challenge!==

Your goal: step by step, recover the original flag.
At each step, you'll see the transformed flag and a hint.
Enter the correct Linux command to reverse the last transformation.

--- Step 1 ---
Current flag: kDQPSdAPGSiNCHMDFnQtpIPIBmYTRLRyInazimLKHmZlonlyRShTRLBicIZj
Hint: base64 encoded the string.
Enter the Linux command to reverse it: base64 -d
Correct!

--- Step 2 ---
Current flag: js93040ss-fa0lgzebsfateG-gk3g-talferirE(SG8pvc
Hint: Reversed the text.
Enter the Linux command to reverse it: rev
Correct!
```

Step 5

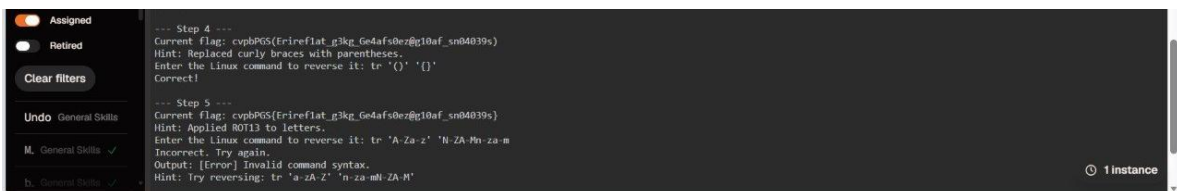
The hint showed that the characters had been changed from underscores to dashes. use the ``tr`` command to change the dashes back into underscores. After I entered the command, the flag changed and the terminal showed **“Correct!”**. So the character replacement was done correctly.



```
--- Step 3 ---
Current flag: cvpbPGS(Erreflat_g3kg-Ge4afs0ez@g10af_sn04039s)
Hint: Replaced underscores with dashes.
Enter the Linux command to reverse it: tr '-' '_'
Correct!
```

Step 6

the hint showed that the parentheses were used instead of the curly braces. use the ``tr`` command to change ``(` and `)`` back to ``{` and `}``. The output changed again, and the terminal showed **“Correct!”**. This means It successfully reversed the Step 4 transformation.

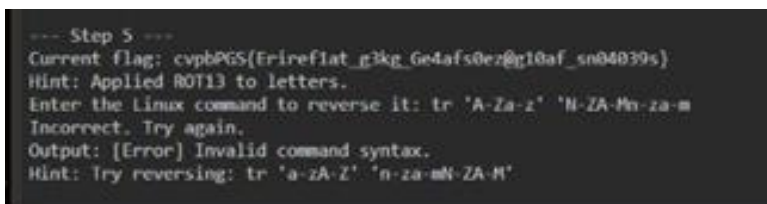


```
--- Step 4 ---
Current flag: cvpbPGS(Erreflat_g3kg-Ge4afs0ez@g10af_sn04039s)
Hint: Replaced curly braces with parentheses.
Enter the Linux command to reverse it: tr `{}` `()'`
Correct!

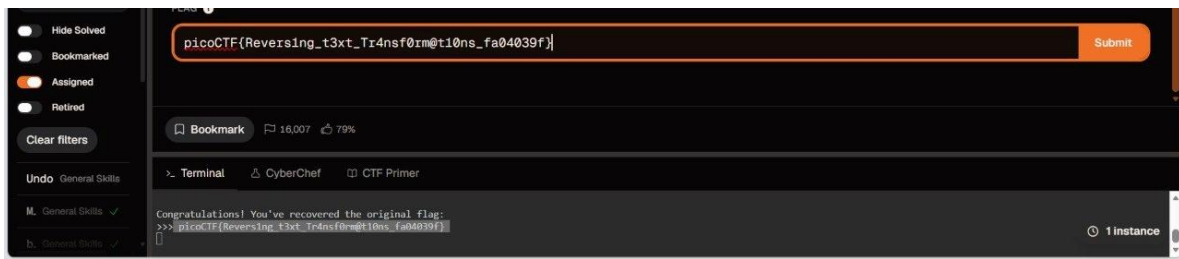
--- Step 5 ---
Current flag: cvpbPGS(Erreflat_g3kg-Ge4afs0ez@g10af_sn04039s)
Hint: Applied ROT13 to letters.
Enter the Linux command to reverse it: tr 'A-Za-z' 'N-ZA-Mn-za-m'
Incorrect. Try again.
Output: [Error] Invalid command syntax.
Hint: Try reversing: tr 'a-zA-Z' 'n-za-mN-ZA-M'
```

Step 7

In Step 5, the letters were changed using **ROT13**. I first tried the command shown in the terminal, but it gave an **“Invalid command syntax”** message. Checked the hint again and corrected the command. After running the correct ``tr`` command, the terminal showed **“Correct!”** and the original flag was recovered.



```
--- Step 5 ---
Current flag: cvpbPGS(Erreflat_g3kg-Ge4afs0ez@g10af_sn04039s)
Hint: Applied ROT13 to letters.
Enter the Linux command to reverse it: tr 'A-Za-z' 'N-ZA-Mn-za-m'
Incorrect. Try again.
Output: [Error] Invalid command syntax.
Hint: Try reversing: tr 'a-zA-Z' 'n-za-mN-ZA-M'
```



5. Flag

picoCTF{Revers1ng_t3xt_Tr4nsf0rm@t10ns_fa04039f}

6. Key Concept / What I Learned

This challenge taught me how Linux text utilities can be used to reverse simple transformations. The important skill was recognizing each transformation from the hint and applying the inverse operation in the correct order, including Base64 decoding, text reversal, character substitution with **tr**, and **ROT13**.

7. References

[picoCTF 2026 | Denton's Blog](#)

YouTube – picoCTF 2026 / CyLab - Undo

https://youtu.be/qTKdqTu7u_A?si=p5XhbHCNesueiOIn

OpenAI. ChatGPT. <https://chatgpt.com/>
